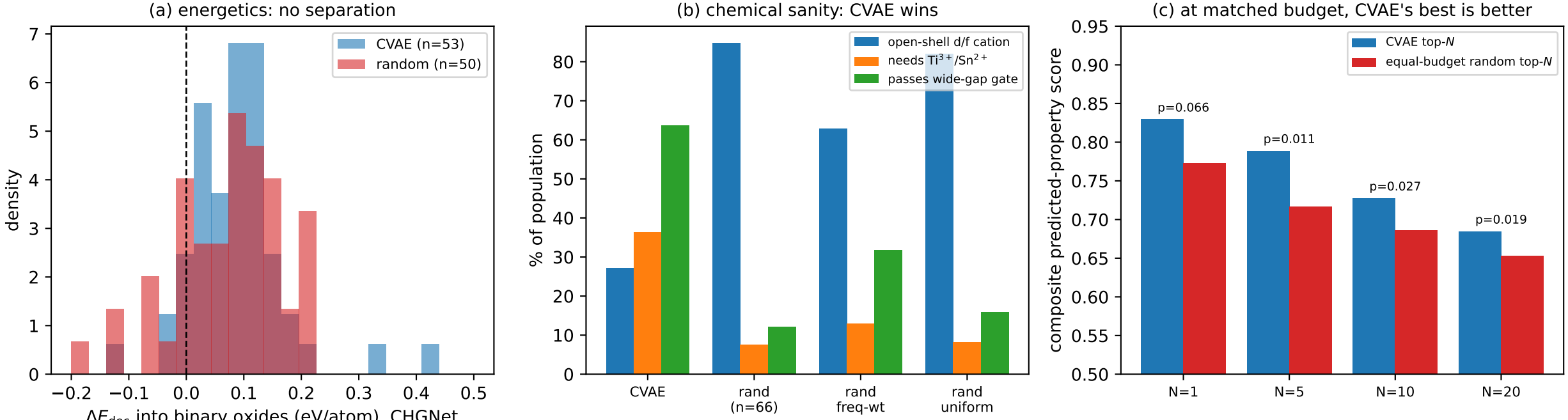
(a) energetics: no separation
CVAE (n=53)
random (n=50)
density
ΔE_{dec} into binary oxides (eV/atom), CHGNet
(b) chemical sanity: CVAE wins
open-shell d/f cation
needs Ti^{3+}/Sn^{2+}
passes wide-gap gate
% of population
CVAE
rand (n=66)
rand freq-wt
rand uniform
(c) at matched budget, CVAE's best is better
CVAE top-N
equal-budget random top-N
composite predicted-property score
p=0.066
p=0.011
p=0.027
p=0.019
N=1
N=5
N=10
N=20